I.P Phishing
Trojan horse.

ck.

AES
(Advance Encryption Standard)

- Block Cipher.
- Symmetric.

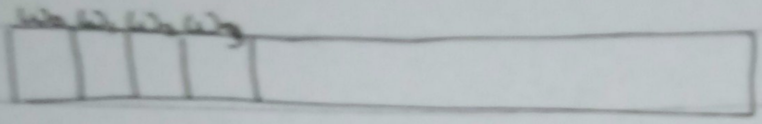
b1	b5	b9	b13
b2	b6	b10	b14
b3	b7	b11	b15
b4	b8	b12	b16
32 bit	32	32	32

= 128 bits

M
128 bits.
state ↓
initial transformation. ← subkey k0.
XOR operation.
state ↓
Round 1. ← subkey 1.
state ↓
Round. ← subkey 2.
⋮
Round 10. ← subkey 10.
|
C.T.

k0
key Expansion

Key expansion.



K_0

initial key

w_0	w_1	w_2	w_3
73	73	6a	72
61	68	73	69
74	63	72	6C
69	6a.	6f	67.

w_4 w_5 w_6 w_7. we want.

$w_3 \rightarrow g \oplus w_0$

$w_4 \oplus w_1$ $w_5 \oplus w_2$ $w_6 \oplus w_3$

Calculate $g(w_3)$.

72 69 6C 67 → shift left.

X1   Y1

w_3	RotWord	Subword	R const	Rconst ⊕ Y1
72	69	F9	01.	F8
69	6C	50	00	50
6C	67	85.	00	85
67	72.	40.	00	40.

$$
\begin{array}{rcccc}
 & 11111001 & 01010000 & 10000101 & 01000000 \\
\oplus & 00000001 & 00000000 & 00000000 & 00000000 \\
\hline
 & 11111000 & 01010000 & 10000101 & 01000000
\end{array}
$$

R constant.

R1	R_2	R_3	R_4	R5
01	02	04	08	10
00				
00				
00				

w_4	w_5	w_6	w_7
8B	F8		
31	59		
F1	92		
29	43		

for w_4.

R const $\oplus\ y_1 \ \oplus\ w_0$.

11111000 01010000 10000101 01000000

$\oplus$ 01~~0~~10101 01100001 01110100 01101001

1000 1011 0011 0001 11110001 0010~~1~~ 1001

8 B 3 1 F 1 2 9

for $w_5 = w_4 \oplus w_1$.

1000 1011 0011 0001 11110001 0010 1001

$\oplus$ 0111 0011 0110 1000 01100011 0110 1010

1111 1000 0101 1001 10010010 0100 0011

F 8 5 9 9 2 ~~8~~4 3.

for $w_6 = w_5 \oplus w_2$.

1111 1000 0101 100~~0~~ 10010010 01000011.

$\oplus$ 0110 1001 0111 0011 01110010 01~~0~~01111.

~~1~~0010001 0010 1010 ~~00~~~~0~~0

0000

0000

00000

Key ~~generation~~ Encryption:

M

128 bits

↓

Add round key.

initial transformation $\oplus$ ← k_0.

← round 0

$k_0 \oplus$ with round key.

↓

Round 1 ← k_1 ($w_4 - w_7$).

↓

Round 2 ← k_2 ($w_8 - w_{11}$).

↓

⋮

Round 10 ← k_{10} ($w_{40} - w_{43}$).

↓

C.T.

Round 1

$\boxed{\text{Byte subsitution}}$

$\boxed{\text{shift row}}$

$\boxed{\text{mix column}}$

$\boxed{\text{Add Round key}}$ ⟷ $k_1(w_4 - w_7)$.

dkey.

k_0.

$-W_7$).

W_8-W_{11}).

$W_{40}-W_{43}$).

Round1

step1 byte subsitution.

00	16	1a	17
04	1C	00	07
17	0e	03	01
1b	0F	0F	10

⇓ see sbox. $0_R 0_C$

63	47	A2	F0
F2	9C	63	C5
F0	Ab	7B	7C
9F	76	76	CA

step2 shift row.

Row 1 = 0-LS
Row2 = 1-LS
Row3 = 2-LS
Row4 = 3-LS

$$\begin{bmatrix} 63 & 47 & A2 & F0 \\ 9C & 63 & C5 & F2 \\ 7B & 7C & F0 & Ab \\ CA & 9F & 76 & 76 \end{bmatrix}$$

Step3 mix column.

costant matrix.

$$\begin{pmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{pmatrix} \times \begin{bmatrix} 63 & 47 & A2 & F0 \\ 9C & 63 & C5 & F2 \\ 7B & 7C & F0 & Ab \\ CA & 9F & 76 & 76 \end{bmatrix} =$$

$$\begin{bmatrix} b1 & b5 & b9 & b13 \\ b2 & b6 & b10 & b14 \\ b3 & b7 & b11 & b15 \\ b4 & b8 & b12 & b16 \end{bmatrix}$$

$r_1 = (02 \times 63) + (03 \times 9C) + (01 \times 7B) + (01 \times CA)$.

$r_2 = (01 \times 63) + (02 \times 9C) + (03 \times 7B) + (01 \times CA)$

for b_1 = (00000010)(01100011) + (00000011)(10011100) +
(00000001)(01111011) + (00000001)(11001010)

GF(2^8) Polynomial Arithmetic $= x^7+x^6+x^5+x^4+x^3+x^2+x+1$.

$= x(x^6+x^5+x+1) + (x+1)(x^7+x^4+x^3+x^2) +$
$1(x^6+x^5+x^4+x^3+x+1) + 1(x^7+x^6+x^3+x)$.

$= x^7+x^6+x^2+x + x^8+x^5+x^4+x^3+x^7+x^4+x^3+x^2+$
$x^6+x^5+x^4+x^3+x+1 + x^7+x^6+x^3+x$.

$x^8+x^7+x^6+x^4+x+1$.
111010011

Irreduceable polynomial.

reduceable polynomial equation.
$P(x) = x^8+x^4+x^3+x+1$.
100011011

step 4.

$$\begin{array}{r|l} \text{P(x):} & \\ 100011011 & 111010011 \\ & \oplus\ 100011011 \\ \hline & 011001000 \\ & \quad B \qquad 8 \end{array}$$

Do for 72.

011100)+
00 1010)

GCD

Q) What is gcd (48, 72)?

72 = 48 × ___ + 8

$72 = 48 \times \underline{1} + \boxed{24}$ gcd.

$48 = 24 \times 2 + 0$

$\underline{\underline{72}} = 48 \times \underline{\ \ } + 24$

What is gcd of (19, 72)?

$72 = 19 \times \underline{3} + 15$

$19 = 15 \times \underline{1} + 4$

$15 = 4 \times \underline{3} + 3$

$4 = 3 \times \underline{1} + \boxed{1}$ gcd

$3 = 1 \times \underline{3} + 0$

Relatively Prime. gcd (19, 72) $\underline{\underline{=1}}$ so Yes imp.

Congruences

$a \cong b \pmod{m}$.

$a \pmod{m} = b$.

Q) Is it true $46 \equiv 68 \pmod{11}$.

$46 \pmod{11} = 68 \pmod{11}$.

$2 = 2$.

yes it is true.

$$11\overline{)46} \quad 4, \quad 46 - 44 = 2.$$

$$11\overline{)68}, \quad 68 - 66 = 2$$

Chinese Remainder theorem.

Q) $x \equiv 2 \pmod{3}$.
$x \equiv 3 \pmod{5}$.
$x \equiv 2 \pmod{7}$.

$x \cong a_1 \pmod{m_1}$.
$x \cong a_2 \pmod{m_2}$.
$x \cong a_3 \pmod{m_3}$.
⋮
$x \cong a_n \pmod{n}$.

foomulae
for x.

$$x = (M_1 X_1 a_1 + M_2 X_2 a_2 + M_3 X_3 a_3 + \ldots\ldots M_n X_n a_n) \bmod$$

for M

$$M = m_1 \times m_2 \times m_3$$

$$M_1 = \frac{M}{m_1} = \frac{\cancel{m_1} \times m_2 \times m_3}{\cancel{m_1}}$$

$M_2 = m_1 \times m_3$ $\quad$ $M_3 = m_1 \times m_2$.

to find x.

$M_1 X_1 \cong 1 \bmod m_1$.

$M_2 X_2 \cong 1 \bmod m_2$.

$M_3 X_3 \cong 1 \bmod m_3$.

if $\gcd(m_1, m_2) = \gcd(m_2\, m_3) = \gcd(m_3\, m_1) = 1$

it is congruence.

Q)

$x \cong \overset{a_1}{1} \pmod{\overset{m_1}{5}}$

$x \cong \overset{a_2}{1} \pmod{\overset{m_2}{7}}$

$x \cong \overset{a_3}{3} \pmod{\overset{m_3}{11}}$

first check gcd.

$\gcd(m_1, m_2) = (5, 7)$

$7 = 5 \times \underline{1} + 2$.

$5 = 2 \times \underline{2} + \boxed{1}$.

$2 = 1 \times \underline{2} + 0$.

$\gcd(m_2\, m_3) = (7, 11)$.

$11 = 7 \times 1 + 4$

$7 = 4 \times 1 + 3$ gcd

$4 = 3 \times 1 + 1$ gcd

$3 = 1 \times \underline{3} + 0$

$\gcd(m_3 m_1) = (11, 5)$.

$11 = 5 \times 2 + 1 \leftarrow$ gcd

$5 = 1 \times 5 + 0$

we can solve further because gcd is 1.

lets cal. M.

$M = 5 \times 7 \times 11.$

$M = 385.$

$M_1 = m_2 \times m_3$

$= 7 \times 11$

$= 77$

$M_2 = 5 \times 11$

$= 55$

$M_3 = 5 \times 7$

$= 35$

Now cal congruence.

$M_1 X_1 \cong 1 \bmod m_1.$

$77 X_1 \cong 1 \bmod \cancel{77}\,5$

$5\overline{)77}$

~~77 X_1 mod 5 = 1 mod 5~~

$77 X_1 \bmod 5 = 1 \bmod 5.$

$5\overline{)6}$ -5 1

$2 X_1 \bmod 5 = 1$

$X_1 = \cancel{8}\,3$

guess X_1

$M_2 X_2 \equiv 1 \bmod m_2$

$55 X_2 \bmod 7 = 1 \bmod 7.$

$6 X_2 \cancel{55} \bmod 7. = 1$

$\cancel{6}\ X_2 = 6.$

$$7 \overline{)55} = 7, \quad 49, \quad 6.$$

$$7 \overline{)36} = 5, \quad 35, \quad 1$$

$M_3 X_3 \cong 1 \bmod m_3.$

$35 X_3 \bmod \cancel{35}\ 11. = 1 \bmod 11.$

$\cancel{35}\ 2 X_3 \bmod 11 = 1.$

$X_3 = 6$

$$11 \overline{)12} = 1, \quad -11, \quad 1.$$

Now calculate X.

$X = ((77 \times 3 \times 1) + (55 \times 6 \times 1) + (35 \times 6 \times 3)) \bmod 385$

$X = \cancel{1422}\ 1191 \bmod 385$

$$385 \overline{)1191}$$

$$\cancel{385 \overline{)1422} = 3, \quad -1155}$$

$\boxed{X = 36}$

$1191 - 36$

Check by putting values in Question.

$36 \cong 1 \pmod 5.$ $\quad 36 \pmod 5 = 1 \bmod 5 \quad 1 = 1.$

$36 \cong 1. \pmod 7.$ $\quad 36 \pmod 7 = 1 \bmod 7$

$36 \cong 3 \pmod{11}.$ $\quad 36 \pmod{11} = 3 \bmod 11$

Q) $x \equiv 2 \pmod{3}$.
$x \equiv 3 \pmod{5}$.
$x \equiv 2 \pmod{7}$

Linear Congruential Generator.

$$x_{n+1} = (a x_n + c) \bmod m.$$
$$x_1 = (a x_0 + c) \bmod m$$
$$x_2 = (a x_1 + c) \bmod m.$$

Period: $\underbrace{7, 17, 23, 1,}_{4\ \text{period}}\ 7, 17, \ldots$

RC4

$$S = \begin{array}{c} \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \\ [\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix}] \end{array},$$

$$T = \begin{array}{c} \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \\ [\begin{matrix} 1 & 2 & 3 & 6 & 1 & 2 & 3 & 6 \end{matrix}] \end{array}$$ - it is k value repeated.

step 2.

$i = 0$
$j = 0$

$$j = (j + S[i] + T[i]) \bmod 8.$$
$$j = (0 + 0 + 1) \bmod 8$$
$$j = 1.$$

swap(S[i], S[j]);

s= [1 0 2 3 4 5 6 7].

j= [1+ 0 +2] mod 8

j= 3

swap(s[i], s[j]).

[1 3 2 0 4 5 6 7].

j= [3+ 2 +0].

j= ~~{~~5

~~swap [S[2], j[5]]~~.

s = [2 3 7 4 6 0 1 5]

last ans.

step 3.

i, j = 0.

while.

i = (i+1) mod 8 ; i=1

j= (j+ S[i]) mod 8.

j= (0 + 3) mod 8 j=3. ⇒ S= [2, 4 7 3 6 0 1 5].

t = (S[i] + S[j]) mod 8;

t= ~~(S~~ 4 + 3 mod 8.

t = 7.

k = S[7] ⇒ k = 5.

k = S[7] ⇒ k=5.

$P = [1\ 2\ 2\ 2\ 2]$.

$k = [5\ \_\ \_\ \_]$.

$C.T = [4\ \_\ \_\ \_]$

for C.T.

$$\begin{array}{r} 001 \\ \oplus\ 101 \\ \hline 100 \\ \hline \end{array}$$ P.T from P.

RSA Algorithm:

Rivest Shamir Adelman.

Step1 Choose any two large Prime no:s P and q.
step = $n = P \times q$.

Step2 $\phi(n) = (P-1)(q-1)$

Step3 choose 'e' public key. $\gcd(n, e) = 1$.

Step4 Calculate Private key $de \cong 1 \bmod n$.
$de \bmod \hat{n} = 1 \bmod n$

Encrypt a plaintext.

$C = M^e \mod n$

To decrypt the block.

$M = C^d \mod n$

$P = 3$ $q = 11$. what is e and d.

Public key Authentication, confidentiality diagram.

d q.

Deffie Hellman key Exchange Algorithm.

1) Choose public numbers: q (large prime no), α (generator mod q).

$q = 7$ α = ? should be Primitive root of q.

$n = \alpha^m \mod q$

$n = 2^1 \mod 7 = 2$

$n = 2^2 \mod 7 = 4$

$2^3 \mod 7 = 1$

$2^4 \mod 7 = 2$

X same so not primitive.

$n = 3^1 \mod 7 = 3$

$= 3^2 \mod 7 = 2$

$3^3 \mod 7 = 6$.

$3^4 \mod 7 = 4$

$3^5 \mod 7 = 5$.

$3^6 \mod 7 = 1$

it is primitive

n

$q = 11$

$\alpha = ?$ it will run till 10 bcz $q = 11$.

$n = \alpha^m \bmod q$.

$2^1 \bmod 11 = 2$.

$2^2 \bmod 11 = 4$

$2^3 \bmod 11 = 8$

$2^4 \bmod 11 = 5$

$2^5 \bmod 11 = 10$

$2^6 \bmod 11 = 9$

$2^7 \bmod 11 = 7$.

$2^8 \bmod 11 =$

$2^9 \bmod 11 =$

$2^{10} \bmod 11 =$

$q = 11 \quad \alpha = 2$.

2) $Y_A = \alpha^{X_A} \bmod q$.

3) $Y_B = \alpha^{X_B} \bmod q$.

2) let $X_A = 4$.

$Y_A = 2^4 \bmod 11 = 16 \bmod 11 = 5$.

3) let $X_B = 6$

$Y_B = 2^6 \bmod 11 = 64 \bmod 11 = 9$.

4) ~~Ex~~cal secret key.

MAC :

1) Symmetric Encryption.
2) Public-key Encryption.

~~1) Symmetric Encryption~~..
~~3~~. Message encryption diagram imp.
Symmetric key encryption (a). } these two diagram.
Public key encryption: (d).

Diagrams message authentication .

* Symetric key encryption using asymmetric.
Distribution of public keys.